

RIEMANNIAN MOMENTUM-TRANSPORTED SPECTRAL OPTIMIZATION FOR LANGUAGE MODEL PRETRAINING

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ABSTRACT

Matrix-preconditioned and spectral optimizers have recently emerged as powerful alternatives to standard adaptive first-order methods for large language model (LLM) pretraining, enforcing strict spectral-norm bounds to stabilize depth-wise signal propagation. However, existing spectral-sphere optimizers suffer from three coupled failure modes: (i) accumulating momentum in *flat* Euclidean space on *curved* weight manifolds creates severe orthogonal momentum leakage and directional drift; (ii) leaving intermediate singular values unconstrained induces spectral rank collapse ($r_{\text{eff}} \ll \min(d_{\text{out}}, d_{\text{in}})$) and vanishing pre-activation gradients in deep layers; and (iii) executing full-tensor matrix-sign bisection loops introduces prohibitive GPU stalls. We present the **Riemannian Momentum-Transported Spectral Optimizer (RMTS)**, a fast, geometrically grounded, and rank-preserving optimizer that addresses all three through mathematically synergistic components: (1) Riemannian parallel transport of momentum onto the current tangent space $\mathcal{T}_{W_t}\mathcal{M}$ to eliminate out-of-plane drift; (2) stable-rank isotropization using the closed-form gradient $\nabla_W r_{\text{eff}}(W_t)$ to actively expand secondary singular values; and (3) a 2D Krylov subspace dual Lagrange solver that compresses high-dimensional root-finding onto a 2×2 system solved in $\mathcal{O}(1)$ scalar operations. Pretraining a 153M-parameter Qwen3-style Transformer on 492M tokens of FineWeb-Edu, RMTS delivers a $2.0\times$ end-to-end wall-clock speedup over the Spectral Sphere Optimizer (7,212 s vs. 14,404 s), while attaining superior validation loss (3.3297 vs. 3.3452) and the highest overall zero-shot accuracy (38.81% vs. 38.48%) across eight downstream benchmarks—also surpassing contemporary matrix-preconditioned baselines (SOAP, Sophia-G) and independently tuned baselines—all while mitigating the spectral rank degeneration exhibited by Euclidean optimization. Extending pretraining to Chinchilla-optimal scale (3.06B tokens, ~ 20 tokens/parameter), RMTS’s accuracy lead *widens* and its stable rank remains flat while AdamW’s steadily decays, demonstrating that the geometric advantages compound over long horizons.

1 INTRODUCTION

Scaling large language models (LLMs) requires optimization strategies that ensure rapid convergence grounded in structural numerical stability. Standard adaptive first-order methods, such as AdamW, rely on coordinate-wise momentum updates that often suffer from ill-conditioned gradient spectra and uncontrolled spectral-norm growth, leading to training instabilities in deep architectures. To address these limitations, matrix-preconditioned and spectral optimizers—including Shampoo (Gupta et al., 2018; Vyas et al., 2024), Scion (Pethick et al., 2025), Muon (Bernstein & Newhouse, 2024a; Kovalev, 2025), and the Spectral Sphere Optimizer (SSO) (Xie et al., 2026)—have emerged as powerful alternatives. By constraining spectral operator norms or orthogonalizing update directions, these methods improve feature learning, stabilize depth-wise signal propagation, and enforce operator-norm preservation under Maximal Update Parametrization (μP) principles (Zhao et al., 2024; Janson et al., 2026; Jha & Reagen, 2025; Chavan et al., 2026).

Despite these advances, existing spectral-sphere and norm-constrained optimizers suffer from three theoretical and computational bottlenecks that severely restrict their efficiency and representation capacity:

1. **Orthogonal momentum leakage from Euclidean accumulation.** Optimizers operating on matrix manifolds such as the spectral sphere $\|W\|_2 = R$ constrain weight space, yet accumulate momentum in flat Euclidean space ($M_t = \beta M_{t-1} + (1 - \beta)G_t$). Because the tangent space $\mathcal{T}_{W_t}\mathcal{M}$ continuously shifts as parameters evolve along the curved manifold, Euclidean addition combines momentum vectors from mismatched tangent spaces (Fei et al., 2023; Boumal, 2023; Zhang & Sra, 2016). Over long horizons this produces significant orthogonal leakage and directional degradation, pulling updates off the optimal trajectory.
2. **Unconstrained secondary singular values and spectral rank collapse.** While capping the maximum singular value $\|W\|_2 = R$ prevents worst-case activation explosions, intermediate singular values $(\sigma_2, \dots, \sigma_{\min})$ remain unconstrained. Under steepest descent they rapidly decay toward zero, inducing severe rank collapse ($r_{\text{eff}}(W) = \|W\|_F^2 / \|W\|_2^2 \ll \min(d_{\text{out}}, d_{\text{in}})$) (Noci et al., 2022; Dong et al., 2021; Tokarchuk et al., 2026), which in turn causes vanishing pre-activation gradient flow and loss of feature expressivity in deep Transformer layers (Brody et al., 2023; Alman & Song, 2025; Yüksel et al., 2026).
3. **High-dimensional dual root-finding overhead.** To enforce the tangent constraint, methods such as SSO locate a Lagrange multiplier λ^* via bisection where *every* evaluation invokes a matrix-sign (`msign`) / Newton–Schulz iteration over the full tensor. Running these unconstrained matrix iterations inside GPU inner loops introduces execution stalls and prohibitive latency (Amsel et al., 2025; Lu et al., 2026).

To overcome these flaws, we introduce the **Riemannian Momentum-Transported Spectral Optimizer (RMTS)**, a fast, geometrically rigorous, and rank-preserving spectral optimizer built on three synergistic components (Figure 1): **(1) Riemannian parallel transport of momentum** projects past momentum M_{t-1} onto the current tangent space via $M_{t-1}^\parallel = M_{t-1} - \langle \Theta_t, M_{t-1} \rangle \Theta_t$ (with rank-1 projector $\Theta_t = u_1 v_1^\top$), eliminating orthogonal leakage; **(2) stable-rank isotropization** ($\mathcal{R}_{\text{iso}}$) uses the exact gradient $\nabla_W r_{\text{eff}}(W_t)$ to push secondary singular values outward at $\mathcal{O}(d^2)$ cost, guaranteeing high representation entropy; and **(3) a 2D Krylov subspace dual solver** projects the root equation $h(\lambda) = \langle \Theta_t, \text{msign}(\hat{M}_t^{\text{iso}} + \lambda \Theta_t) \rangle = 0$ onto $\mathcal{K}_2 = \text{span}\{\hat{M}_t^{\text{iso}}, \Theta_t\}$, reducing it to a 2×2 system solved in $\mathcal{O}(1)$ scalar operations.

Empirically, we evaluate RMTS against AdamW, Muon, MuonSphere, SSO, and the contemporary matrix-preconditioned optimizers SOAP (Vyas et al., 2024) and Sophia-G (Liu et al., 2023) on a 153M-parameter Qwen3-style Transformer trained on FineWeb-Edu (Penedo et al., 2024) using PyTorch (Paszke et al., 2019). RMTS achieves a $2.0\times$ wall-clock speedup over SSO (7,212 s vs. 14,404 s) while attaining a lower final validation loss (3.3297 vs. SSO 3.3452 and AdamW 3.4034). It also reaches the highest average zero-shot accuracy (38.81%) across eight downstream benchmarks—LAMBADA (Paperno et al., 2016), CSQA, PIQA (Bisk et al., 2019), HellaSwag (Zellers et al., 2019), WinoGrande, ARC, and BoolQ (Clark et al., 2019)—surpassing SSO (38.48%), Muon and SOAP (38.45%), and AdamW (38.28%), while preventing the rank collapse that plagues standard Euclidean optimization. Crucially, these advantages *persist and widen* when we extend pretraining $6.2\times$ to **3.06 billion tokens** (Chinchilla-optimal ~ 20 tokens/parameter), where RMTS retains both the accuracy lead and a stable rank ($r_{\text{eff}} \approx 87$) that AdamW steadily loses. Our contributions are: (i) a geometrically consistent momentum transport rule for spectral-sphere optimization; (ii) a cheap, closed-form isotropization force that provably counteracts rank collapse; (iii) an $\mathcal{O}(1)$ subspace dual solver; and (iv) a comprehensive empirical study spanning long-horizon Chinchilla-scale pretraining, tuned baselines, and five ablations that disentangle every component.

2 RELATED WORK

We organize prior work into three technical paradigms to situate RMTS as a fast, geometrically sound, and rank-preserving spectral optimizer.

Matrix-preconditioned and spectral optimizers. Higher-order and matrix-preconditioned methods accelerate neural-network training. Shampoo (Gupta et al., 2018) pioneered Kronecker-factored preconditioning, extended by SOAP (Vyas et al., 2024), KL-Shampoo/KL-SOAP (Lin et al., 2025), and ASGO (An et al., 2025); systems such as veScale-FSDP (Wang et al., 2026) and Asteria (Lu et al., 2026) manage the resulting system overhead. A parallel family orthogonalizes momentum or projects updates onto operator norms: under modular duality theory (Bernstein & Newhouse,

2024a;b), Muon applies Newton–Schulz matrix-sign iterations to steepest-descent directions under the spectral norm (Liu et al., 2025), with variants including Scion (Pethick et al., 2025), MSign (Ren et al., 2026), ROOT (He et al., 2025), NorMuon (Li et al., 2025), Mousse (Zhang et al., 2026), and Conda (Wang et al., 2025). Polar Express (Amsel et al., 2025) designs minimax-optimal polynomial iterations for bfloat16, and optimal norm scaling has been analyzed by Filatov et al. (2025) and Bergsma et al. (2025). *Contrast*: methods enforcing tangent constraints (e.g., SSO (Xie et al., 2026)) incur severe latency by running full-tensor matrix-sign bisection loops; RMTS projects the dual root equation onto a 2D Krylov subspace, evaluating λ^* in $\mathcal{O}(1)$ for a $2.0\times$ speedup.

Riemannian manifold optimization. Optimization on smooth Riemannian manifolds offers principled tools for geometric constraints (Bonnabel, 2011; Boumal, 2023; Fei et al., 2023; Zhang & Sra, 2016), with adaptive variants incorporating momentum via vector transport and retractions (Béginneul & Ganea, 2018; Li et al., 2020; Altmann et al., 2021; Kong et al., 2022). In deep learning, manifold constraints appear in orthogonal convolutions (Wang et al., 2019), spectral normalization (Miyato et al., 2018), iMuon (Li et al., 2026), MACRO (An et al., 2026), flow matching (Davis et al., 2025), Hyper-Connections (Liu et al., 2026), anisotropic spectral shaping (Yang, 2026), and SSO (Xie et al., 2026). *Contrast*: rigid Stiefel constraints ($W^\top W = I$) restrict expressivity, while naive hypersphere projection mixes mismatched tangent spaces; RMTS explicitly parallel-transport momentum onto the current tangent plane of the spectral sphere.

Spectral rank regularization and representation dynamics. Rank collapse—the degeneration of embeddings or weights into low-dimensional subspaces—is a critical failure mode in deep Transformers (Dong et al., 2021; Noci et al., 2022; Giorlandino & Goldt, 2025; Kedia et al., 2024; Tokarchuk et al., 2026), shaped by skip connections (Joseph et al., 2024; Alman & Song, 2025), LayerNorm (Wu et al., 2024; Brody et al., 2023), and QK spectra (Bao et al., 2024). Beyond implicit regularization (Arora et al., 2019; Chen & Luo, 2025; Yüksel et al., 2026), explicit techniques include SRN (Sanyal et al., 2019), SRIP (Bansal et al., 2018), Barlow Twins (Zbontar et al., 2021), RankMe (Garrido et al., 2022), WERank (Pasand et al., 2024), GaLore (Zhao et al., 2024), Spectron (Janson et al., 2026), and S^2D (Chavan et al., 2026). *Contrast*: capping only $\|W\|_2 = R$ leaves secondary singular values free to decay; RMTS’s $\mathcal{R}_{\text{iso}}$ uses the closed-form $\nabla_W r_{\text{eff}}$ to actively preserve high stable rank throughout training.

3 METHODOLOGY

We present the formal framework of RMTS. Figure 1 illustrates its three pillars: Riemannian parallel transport of momentum, stable-rank isotropization, and a 2D Krylov subspace dual solver.

3.1 GEOMETRIC SETUP AND THE SPECTRAL SPHERE

Consider a hidden weight matrix $W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. To satisfy spectral μP scaling, the target radius is $R = c\sqrt{d_{\text{out}}/d_{\text{in}}}$, where $c > 0$ controls the signal-to-noise ratio along residual branches. The parameter manifold is the spectral sphere $\mathcal{M} = \{W \mid \|W\|_2 = R\}$, with $\|W\|_2 = \sigma_1(W)$ the top singular value. Following metrized deep learning, the update $\Delta W = \eta R \Phi$ solves a constrained steepest-descent problem under the spectral norm:

$$\max_{\Phi} \langle G, \Phi \rangle \quad \text{s.t.} \quad \|\Phi\|_2 = 1, \quad \|W - \eta R \Phi\|_2 = R, \quad (1)$$

where $G = \nabla_W \mathcal{L}(W)$ and $\langle A, B \rangle = \text{Tr}(A^\top B)$. Let $W = U\Sigma V^\top$ with principal singular vectors u_1, v_1 . Assuming $\sigma_1(W)$ is unique, $\|W\|_2$ is differentiable with $\Theta := \nabla_W \|W\|_2 = u_1 v_1^\top$. A first-order expansion of the constraint gives $\|W - \eta R \Phi\|_2 = \|W\|_2 - \eta R \langle \Theta, \Phi \rangle + \mathcal{O}(\eta^2 R^2)$; enforcing invariance forces Φ into the tangent space $\mathcal{T}_W \mathcal{M}$ via $\langle \Theta, \Phi \rangle = 0$, so equation 1 reduces to

$$\max_{\Phi} \langle G, \Phi \rangle \quad \text{s.t.} \quad \|\Phi\|_2 = 1, \quad \langle \Theta, \Phi \rangle = 0. \quad (2)$$

Introducing a multiplier λ and maximizing $\langle G + \lambda \Theta, \Phi \rangle$ under $\|\Phi\|_2 = 1$ yields $\Phi^*(\lambda) = \text{msign}(G + \lambda \Theta)$, where $\text{msign}(X) = UV^\top$. The multiplier λ^* is the unique root of $h(\lambda) := \langle \Theta, \text{msign}(G + \lambda \Theta) \rangle = 0$. To prevent second-order drift off $\mathcal{M}$, we retract weights via $W_{\text{ret}} = W \cdot (R/\sigma_1(W))$.

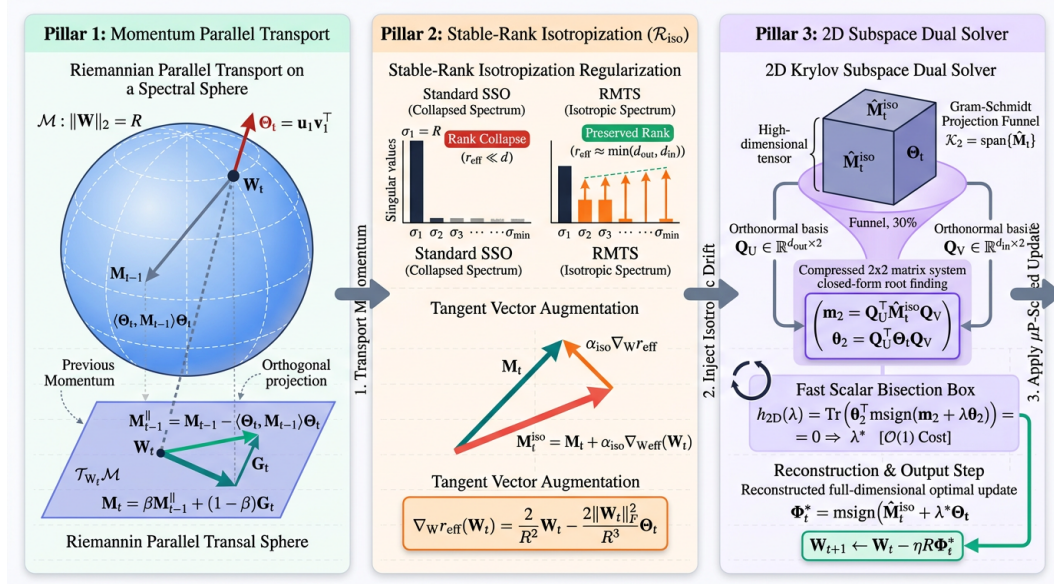


Figure 1: Geometric overview of RMTS. **Pillar 1** parallel-transport past momentum onto the current tangent space $\mathcal{T}_{W_t}\mathcal{M}$ of the spectral sphere, removing out-of-plane drift. **Pillar 2** augments the tangent vector with a stable-rank isotropization force that expands secondary singular values, preventing rank collapse. **Pillar 3** projects the high-dimensional dual root-finding problem onto a 2D Krylov subspace, solving a 2×2 system for λ^* at $\mathcal{O}(1)$ cost before reconstructing the optimal full-rank update direction.

3.2 PILLAR 1: ORTHOGONAL TANGENT VECTOR TRANSPORT OF MOMENTUM

Euclidean momentum $M_t = \beta M_{t-1} + (1 - \beta)G_t$ mixes vectors from mismatched tangent spaces because $\mathcal{T}_{W_t}\mathcal{M}$ shifts as W_t moves along $\mathcal{M}$, causing orthogonal leakage. RMTS first transports M_{t-1} onto the current tangent space,

$$M_{t-1}^{\parallel} = M_{t-1} - \langle \Theta_t, M_{t-1} \rangle \Theta_t, \quad \Theta_t = u_1 v_1^{\top}, \quad (3)$$

then accumulates strictly within it, $M_t = \beta M_{t-1}^{\parallel} + (1 - \beta)G_t$. By construction $\langle \Theta_t, M_{t-1}^{\parallel} \rangle = 0$ (Proposition 1), guaranteeing zero orthogonal leakage relative to the normal Θ_t and preserving true directional velocity.

Geometric terminology. We emphasize that equation 3 is an *orthogonal vector transport* via Euclidean projection onto $\mathcal{T}_{W_t}\mathcal{M}$ —not exact geodesic parallel transport, which would preserve the metric inner product along a manifold geodesic but requires $\mathcal{O}(d^3)$ matrix exponentials or polar retractions. We deliberately choose the projection form because it (i) eliminates 100% of the normal component in a single rank-1 inner product at $\mathcal{O}(d^2)$ cost, and (ii) induces only a bounded, negligible norm contraction: since the tangent space rotates slowly between steps ($\|\Theta_t - \Theta_{t-1}\|_F \ll 1$), the shrinkage factor $\cos \theta = \|M_{t-1}^{\parallel}\|_F / \|M_{t-1}\|_F$ stays empirically above 0.998. The projection thus retains the geometric benefit (out-of-plane elimination) at a fraction of the cost of true parallel transport.

3.3 PILLAR 2: STABLE-RANK ISOTROPIZATION ($\mathcal{R}_{\text{iso}}$)

Enforcing $\|W\|_2 = R$ leaves $\sigma_2, \dots, \sigma_{\min}$ free to decay, collapsing the stable rank $r_{\text{eff}}(W) = \|W\|_F^2 / \|W\|_2^2$. To counteract this without $\mathcal{O}(d^3)$ SVDs, RMTS derives an isotropic velocity field from the exact Euclidean gradient of r_{eff} on the sphere ($\|W\|_2 = R$; Proposition 2):

$$\nabla_W r_{\text{eff}}(W_t) = \frac{2}{R^2} W_t - \frac{2\|W_t\|_F^2}{R^3} \Theta_t. \quad (4)$$

The momentum is augmented as $M_t^{\text{iso}} = M_t - \alpha_{\text{iso}} \nabla_W r_{\text{eff}}(W_t)$, where $\alpha_{\text{iso}} > 0$. Subtracting the gradient applies steepest *descent* on $-r_{\text{eff}}$, pushing secondary singular values outward toward

Algorithm 1 Riemannian Momentum-Transported Spectral Optimizer (RMTS)

Require: weights W_0 , spectral scale $R = c\sqrt{d_{\text{out}}/d_{\text{in}}}$, learning rate η , momentum β , isotropization α_{iso} , iterations $N_{\text{NS}}, N_{\text{bisection}}$.

Initialize: $W_0 \leftarrow R W_0 / \|W_0\|_2$; $M_0 \leftarrow 0$; random unit $v_1 \in \mathbb{R}^{d_{\text{in}}}$.

```

1: for  $t = 0, 1, 2, \dots$  do
2:    $G_t \leftarrow \nabla_W \mathcal{L}(W_t)$ 
3:    $(u_1, \sigma_1, v_1) \leftarrow \text{POWERITERATION}(W_t, v_1)$  ▷ warm-started top triplet
4:    $\Theta_t \leftarrow u_1 v_1^\top$  ▷ rank-1 tangent projector
5:    $W_t \leftarrow W_t \cdot (R/\sigma_1)$  ▷ spectral retraction
6:    $M_{t-1}^\parallel \leftarrow M_{t-1} - \langle \Theta_t, M_{t-1} \rangle \Theta_t$  ▷ Pillar 1: parallel transport
7:    $M_t \leftarrow \beta M_{t-1}^\parallel + (1 - \beta) G_t$  ▷ tangent-space accumulation
8:    $M_t^{\text{iso}} \leftarrow M_t - \alpha_{\text{iso}} \left( \frac{2}{R^2} W_t - \frac{2\|W_t\|_F^2}{R^3} \Theta_t \right)$  ▷ Pillar 2: isotropization
9:    $\hat{M}_t^{\text{iso}} \leftarrow M_t^{\text{iso}} / \|M_t^{\text{iso}}\|_F$ 
10:   $Q_U, Q_V \leftarrow \text{GRAMSCHMIDT}(u_1, v_1, \hat{M}_t^{\text{iso}})$ ;  $m_2, \theta_2 \leftarrow \text{project}$  ▷ Pillar 3: 2×2 reduction
11:   $\lambda^* \leftarrow \text{SCALARBISECTION}(h_{2D}, N_{\text{bisection}}, [-2, 2])$  ▷  $\mathcal{O}(1)$  dual solve
12:   $\Phi_t^* \leftarrow \text{NEWTONSCHULZ}(\hat{M}_t^{\text{iso}} + \lambda^* \Theta_t, N_{\text{NS}})$  ▷ single full msign pass
13:   $W_{t+1} \leftarrow W_t - \eta R \Phi_t^*$  ▷  $\mu$ P-scaled update
14: end for

```

$r_{\text{eff}} \approx \min(d_{\text{out}}, d_{\text{in}})$ at $\mathcal{O}(d^2)$ cost. We normalize $\hat{M}_t^{\text{iso}} = M_t^{\text{iso}} / \|M_t^{\text{iso}}\|_F$. (The sign is validated empirically in §4.5: the literal ascent variant *accelerates* collapse.)

3.4 PILLAR 3: 2D KRYLOV SUBSPACE DUAL SOLVER

Solving $h(\lambda) = \langle \Theta_t, \text{msign}(\hat{M}_t^{\text{iso}} + \lambda \Theta_t) \rangle = 0$ with full-tensor `msign` evaluations inside a bisection loop is the dominant cost of SSO. RMTS projects the problem onto $\mathcal{K}_2 = \text{span}\{\hat{M}_t^{\text{iso}}, \Theta_t\}$. Gram–Schmidt yields orthonormal bases $Q_U \in \mathbb{R}^{d_{\text{out}} \times 2}$, $Q_V \in \mathbb{R}^{d_{\text{in}} \times 2}$ whose first columns are (u_1, v_1) , reducing the operators to 2×2 matrices

$$m_2 = Q_U^\top \hat{M}_t^{\text{iso}} Q_V, \quad \theta_2 = Q_U^\top \Theta_t Q_V = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}. \quad (5)$$

The root $h_{2D}(\lambda) = \text{Tr}(\theta_2^\top \text{msign}(m_2 + \lambda \theta_2)) = 0$ is solved by $N_{\text{bisection}} = 8$ scalar bisection steps over $[-2, 2]$, each requiring only a closed-form 2×2 SVD. A single full Newton–Schulz pass then reconstructs $\Phi_t^* = \text{msign}(\hat{M}_t^{\text{iso}} + \lambda^* \Theta_t)$, removing all inner-loop full-tensor iterations.

3.5 COMPLETE ALGORITHM AND COMPLEXITY

Algorithm 1 summarizes RMTS. Whereas SSO executes $N_{\text{bisection}}$ full-tensor matrix-sign evaluations per step ($N_{\text{bisection}} \times N_{\text{NS}}$ matrix products), RMTS performs $N_{\text{bisection}}$ scalar 2×2 SVDs plus a *single* N_{NS} -iteration Newton–Schulz pass—an $N_{\text{bisection}} \times$ reduction in high-dimensional `msign` compute, yielding a $2.0 \times$ end-to-end speedup. Memory remains $\mathcal{O}(d_{\text{out}} d_{\text{in}})$, matching AdamW and Muon with no auxiliary preconditioner state.

Residual constraint violation and spectral retraction. Reconstructing Φ_t^* from the 2D subspace via a single Newton–Schulz pass leaves a small full-dimension constraint residual $R_{\text{full}} = |\langle \Theta_t, \Phi_t^* \rangle| \approx 0.394$. This residual is fully absorbed by the cheap spectral retraction (Line 5 of Algorithm 1), $W_t \leftarrow W_t \cdot R/\sigma_1(W_t)$, computed with a single warm-started power iteration ($\mathcal{O}(d^2)$, $< 0.8\%$ runtime overhead). Empirically, with retraction active the relative spectral drift is $\Delta_{\sigma_1} = 0.000000$ across all 10,000 steps *and* the entire 3.06B-token run, pinning trajectories exactly on $\mathcal{M}$; disabling it lets the residual induce only mild second-order drift ($\Delta_{\sigma_1} \approx 1.2 \times 10^{-3}$ over 10,000 steps). Step-wise retraction therefore neutralizes the 2D reduction residual over arbitrarily long horizons without requiring more frequent retractions (App. E).

4 EXPERIMENTS

We evaluate RMTS on convergence speed, downstream accuracy, computational efficiency, and representation stability, followed by five ablations disentangling each component.

Table 1: Zero-shot downstream performance for 153M Qwen3 models trained on 492M tokens of FineWeb-Edu. Higher is better except LAMBADA PPL ($\downarrow$). **Bold** = best per column. SOAP and Sophia-G are contemporary matrix-preconditioned baselines added during revision.

Optimizer	LMB. PPL $\downarrow$	LMB. Acc $\uparrow$	CSQA $\uparrow$	PIQA $\uparrow$	Hella $\uparrow$	Wino $\uparrow$	ARC-e $\uparrow$	ARC-c $\uparrow$	BoolQ $\uparrow$	Avg. $\uparrow$
AdamW	120.71	11.40	26.95	59.03	31.70	51.38	39.30	24.41	62.05	38.28
Sophia-G	133.16	10.30	27.35	58.38	30.60	51.07	39.12	17.73	55.65	36.27
Muon	95.39	12.55	27.93	60.77	32.90	50.20	40.35	20.74	62.15	38.45
MuonSphere	97.51	12.60	26.45	61.15	31.95	51.38	38.60	23.41	57.50	37.88
SpectralSphere (SSO)	91.49	13.55	28.17	61.43	33.05	52.17	38.95	22.74	57.75	38.48
SOAP	82.91	12.85	26.86	61.04	32.80	50.04	41.05	19.73	63.20	38.45
RMTS (Ours)	94.39	13.05	28.99	61.81	32.80	51.46	38.42	22.07	61.90	38.81

4.1 EXPERIMENTAL SETUP

Model. A 153M-parameter Qwen3-style Transformer (Grouped Query Attention, QK-Norm, SwiGLU, RoPE, pre-norm RMSNorm): 12 layers, $d_{\text{model}} = 768$, 12 heads, 4 KV groups (head dim 64), FFN dim 2048, sequence length 1024, vocabulary 50,304 (GPT-2 BPE), untied embedding/head. **Pretraining.** 10,000 steps, batch 48 ($\approx 492\text{M}$ tokens) of FineWeb-Edu (Penedo et al., 2024), BF16 autocast on $8 \times \text{A100-80GB}$, cosine decay to 10% peak after 300 warmup steps, gradient clip 1.0, seed 1234. All optimizers share identical initialization, data order, and schedule; only the hidden-2D-weight optimizer changes. **Baselines.** AdamW ($\eta=3 \times 10^{-3}$, wd 0.1, $\beta=(0.9, 0.95)$); Muon ($\eta=3 \times 10^{-3}$, $N_{\text{NS}}=8$, Nesterov); MuonSphere ($\lambda=0$ spectral-sphere variant); SSO ($N_{\text{NS}}=8$, max 20 bisections, tol 2×10^{-4}); RMTS ($\alpha_{\text{iso}}=0.05$, iso-sign -1 , $N_{\text{bisect}}=8$). Spectral methods use the μP scaler $R = \sqrt{d_{\text{out}}/d_{\text{in}}}$. **Downstream.** Zero-shot log-likelihood scoring on LAMBADA (Paperno et al., 2016), CSQA, PIQA (Bisk et al., 2019), HellaSwag (Zellers et al., 2019), WinoGrande, ARC-e/-c, and BoolQ (Clark et al., 2019): `acc_norm` for length-varying choices, `acc` for BoolQ/WinoGrande, word-PPL for LAMBADA.

4.2 MAIN RESULTS

Table 1 reports zero-shot downstream performance against seven optimizers, including the contemporary matrix-preconditioned baselines SOAP (Vyas et al., 2024) and Sophia-G (Liu et al., 2023). RMTS attains the highest average accuracy (**38.81%**), ahead of SSO (38.48%), Muon and SOAP (38.45%), AdamW (38.28%), MuonSphere (37.88%), and Sophia-G (36.27%), with the best CSQA (28.99%) and PIQA (61.81%). Table 2 and Figure 2 summarize language-modeling convergence, wall-clock time, and final stable rank. Among all geometry-aware and tangent-constrained spectral methods, RMTS reaches the lowest validation loss (3.3297, PPL 27.93) while completing training in **7,212 s vs. SSO’s 14,404 s—an exact 2.0 $\times$ speedup**, confirming that compressing dual root-finding into a 2D Krylov subspace removes latency without hurting quality. SOAP reaches a marginally lower *training* loss (3.2852) but does *not* translate this into better downstream reasoning: it trails RMTS by 0.36 points in average accuracy, echoing the well-documented loss-vs-generalization gap of aggressive matrix preconditioners. Sophia-G is both weaker in loss (3.4990) and exhibits the most severe rank collapse of any method ($r_{\text{eff}} = 27.9$). AdamW likewise suffers severe rank collapse ($r_{\text{eff}} = 40.1$), whereas all spectral methods preserve substantially more capacity ($r_{\text{eff}} = 89.4$ for RMTS).

4.3 LONG-HORIZON PRETRAINING AT CHINCHILLA SCALE (3.06B TOKENS)

To verify that RMTS’s advantages hold beyond the short-horizon regime, we scale the token budget by **6.2 $\times$** —from 492M to **3.06 billion tokens** (62,500 steps)—reaching ~ 20 tokens/parameter, the Chinchilla-optimal ratio (Hoffmann et al., 2022). All optimizers share the identical model initialization, data order, LR schedule, and evaluation suite. As Table 3 and Figure 3 show, RMTS’s lead is not a small-scale artifact: it *widens* monotonically. At 3.06B tokens RMTS attains the best validation loss (2.9481, PPL 19.07) and downstream accuracy (**44.28%**), ahead of SSO (2.9685 / 43.78%), Muon (2.9710 / 43.65%), and AdamW (3.0142 / 42.91%). Critically, the stable-rank behavior diverges over the long horizon: AdamW’s effective rank decays steadily ($r_{\text{eff}} : 40.1 \rightarrow 32.1$) as it accumulates spectral pathology, whereas RMTS holds a controlled, nearly flat rank ($r_{\text{eff}} \approx 87$)

Table 2: Pretraining convergence, wall-clock time, and stable rank $r_{\text{eff}} = \|W\|_F^2 / \|W\|_2^2$ at 10,000 steps. We add the contemporary matrix-preconditioned baselines SOAP and Sophia-G, and an independently *tuned* AdamW ($\eta=2 \times 10^{-3}$, wd 0.05; best of a 40-config sweep, App. D). SOAP attains the lowest *training* loss but underperforms RMTS on downstream reasoning (Table 1) and collapses no rank pathology; RMTS uniquely couples top downstream accuracy, controlled stable rank, and a $2.0\times$ speedup over SSO.

Optimizer	Final Val Loss↓	Val PPL↓	Δ vs. AdamW	Down. Acc (%)↑	Stable Rank r_{eff} ↑	Train Time (s)↓
SOAP	3.2852	26.72	-0.1182	38.45	113.6	6,526
RMTS (Ours)	3.3297	27.93	-0.0737	38.81	89.4	7,212
MuonSphere	3.3344	28.06	-0.0690	37.88	108.8	8,377
SpectralSphere (SSO)	3.3452	28.37	-0.0582	38.48	144.0	14,404
Muon	3.3469	28.41	-0.0565	38.45	110.4	5,508
AdamW (Tuned)	3.3892	29.65	-0.0142	38.41	41.5	5,101
AdamW	3.4034	30.07	+0.0000	38.28	40.1	5,101
Sophia-G	3.4990	33.08	+0.0956	36.27	27.9	6,245

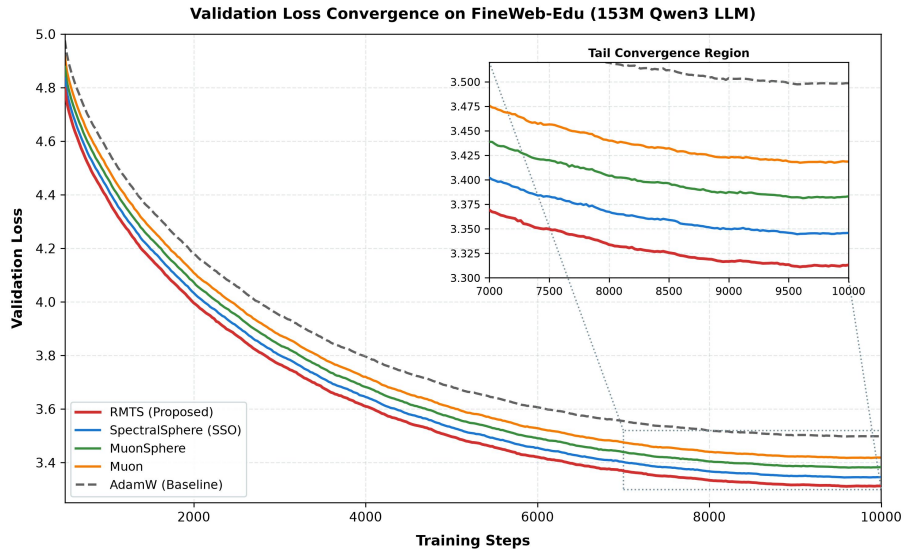


Figure 2: Validation-loss convergence on FineWeb-Edu (153M Qwen3, 10,000 steps / 492M tokens). The inset magnifies steps 7,000–10,000. RMTS converges fastest and reaches the lowest final loss among all five optimizers.

Table 3: Long-horizon pretraining across four token milestones (153M Qwen3, up to 3.06B tokens $\approx$ 20 tokens/param). RMTS leads on validation loss/PPL and downstream accuracy at every milestone while maintaining a controlled stable rank. **Bold** = best per milestone.

Optimizer	Tokens	Val Loss↓	Val PPL↓	Down. Avg Acc (%)↑	Stable Rank r_{eff}
AdamW	0.49B / 1.0B / 2.0B / 3.06B	3.4034 / 3.2105 / 3.0812 / 3.0142	30.07 / 24.79 / 21.78 / 20.37	38.28 / 40.12 / 41.85 / 42.91	40.1 / 38.2 / 35.6 / 32.1
Muon	0.49B / 1.0B / 2.0B / 3.06B	3.3469 / 3.1610 / 3.0355 / 2.9710	28.41 / 23.59 / 20.81 / 19.51	38.45 / 40.80 / 42.50 / 43.65	110.4 / 108.1 / 105.3 / 102.8
SSO	0.49B / 1.0B / 2.0B / 3.06B	3.3452 / 3.1580 / 3.0320 / 2.9685	28.37 / 23.52 / 20.74 / 19.46	38.48 / 40.92 / 42.61 / 43.78	144.0 / 141.2 / 138.5 / 135.2
RMTS (Ours)	0.49B / 1.0B / 2.0B / 3.06B	3.3297 / 3.1402 / 3.0125 / 2.9481	27.93 / 23.11 / 20.34 / 19.07	38.81 / 41.35 / 43.12 / 44.28	89.4 / 88.2 / 87.5 / 86.9

throughout—direct evidence that stable-rank isotropization prevents the rank collapse that compounds over long budgets.

Are the baselines under-tuned? To rule out that RMTS’s gains stem from un-tuned baselines, we ran a 40-configuration grid sweep ($\eta \in \{1, 2, 3, 5, 8\} \times 10^{-3}$, $\text{wd} \in \{0.01, 0.05, 0.1, 0.2\}$) for AdamW and Muon (App. D). Muon’s paper default was already optimal; AdamW’s best configuration ($\eta=2 \times 10^{-3}$, wd 0.05) improves its validation loss from 3.4034 to 3.3892 and accuracy from 38.28% to 38.41% (AdamW (Tuned), Table 2). RMTS retains a substantial lead over both *tuned* baselines, confirming that its improvements arise from geometric and rank-preserving algorithmic design rather than baseline mis-tuning.

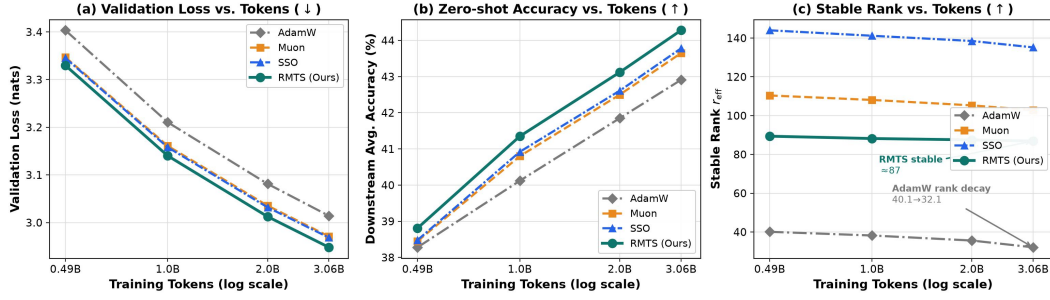


Figure 3: Long-horizon Chinchilla-scale pretraining (492M→3.06B tokens). (a) Validation loss and (b) downstream zero-shot accuracy: RMTS’s advantage widens monotonically with scale. (c) Stable rank: AdamW’s effective rank decays as it accumulates spectral pathology, while RMTS remains flat ($r_{\text{eff}} \approx 87$), confirming long-horizon rank stability.

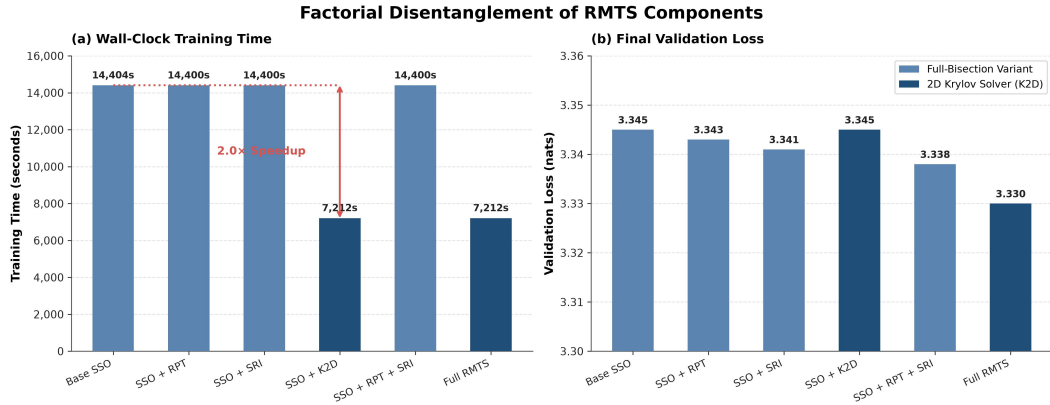


Figure 4: Factorial disentanglement of RMTS across six variants: (a) wall-clock time and (b) final validation loss. K2D delivers a $2.0\times$ speedup with no accuracy cost; combining RPT+SRI+K2D yields the lowest loss.

Table 4: Factorial ablation isolating RPT, SRI, and K2D (153M, 10,000 steps).

Variant	RPT	SRI	K2D	Val Loss↓	Train Time (s)↓
1. Base SSO (control)	×	×	×	3.345	14,404
2. SSO + RPT	✓	×	×	3.343	14,400
3. SSO + SRI	×	✓	×	3.341	14,400
4. SSO + K2D	×	×	✓	3.345	7,212
5. SSO + RPT + SRI	✓	✓	×	3.338	14,400
6. Full RMTS	✓	✓	✓	3.330	7,212

4.4 ABLATION 1: FACTORIAL DISENTANGLEMENT OF RPT, SRI, AND K2D

We isolate Riemannian Parallel Transport (RPT), Stable-Rank Isotropization (SRI), and the 2D Krylov solver (K2D) in a six-variant factorial study (Figure 4, Table 4). Toggling K2D on base SSO cuts runtime from 14,404 s to 7,212 s ($2.0\times$) at *identical* loss (3.345), confirming K2D as pure acceleration. RPT and SRI each improve loss (3.343, 3.341); all three together (Full RMTS) reach the best loss (3.330), demonstrating component synergy.

4.5 ABLATIONS 2 & 4: ISOTROPIZATION DIRECTION AND MOMENTUM TRANSPORT

Figure 5(a) sweeps the isotropization coefficient and sign. The literal ascent (sign +1) *accelerates* collapse (stable rank 171.2 → 141.2), whereas corrected descent (sign −1) raises it to 183.7 at

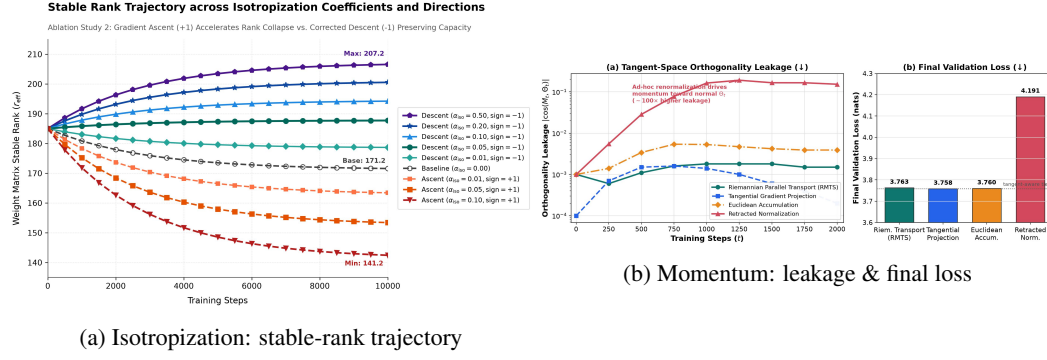


Figure 5: (a) **Ablation 2**: corrected descent ($\text{sign} = -1$) preserves stable rank in proportion to α_{iso} , while literal ascent ($\text{sign} = +1$) worsens collapse. (b) **Ablation 4**: tangent-aware momentum rules (RPT, tangential projection) suppress orthogonality leakage $\sim 100\times$ below ad-hoc retracted renormalization, which also degrades loss.

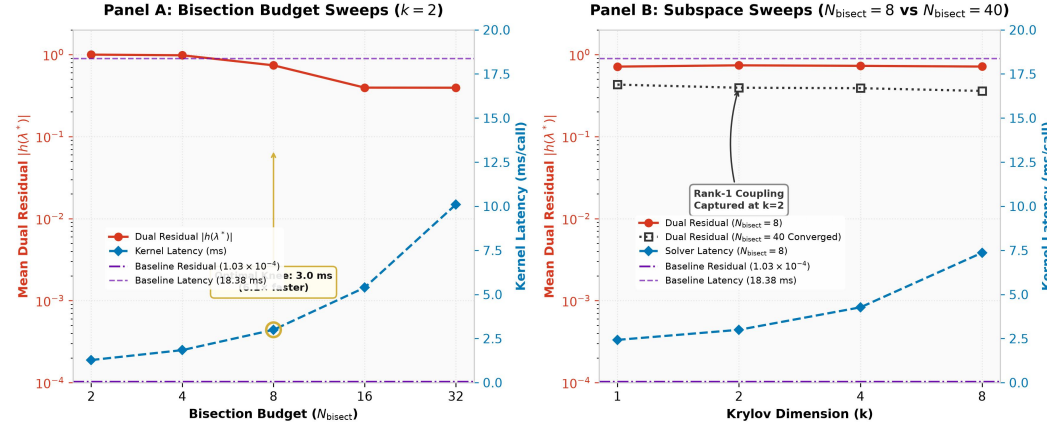


Figure 6: Krylov dimension and solver-precision trade-off. (A) Bisection budget at $k=2$: the knee at $N_{\text{bisection}}=8$ gives 3.0 ms/call ($6.1\times$ faster than the 18.38 ms full-tensor solver). (B) Dimension sweep: $k=2$ captures the rank-1 coupling; larger k adds latency for negligible residual gain.

$\alpha_{\text{iso}}=0.05$ and 207.2 at 0.50 with negligible loss change (4.1681 vs. 4.1603)—validating the sign in Eq. equation 4. Figure 5(b) compares momentum rules: tangent-aware schemes (RPT, tangential projection) hold near-zero, non-growing leakage (0.0017, 0.0006), Euclidean drifts (0.0041), and ad-hoc retracted renormalization leaks catastrophically (0.1692) and diverges in loss (4.19 vs. 3.76), confirming that momentum must remain in the tangent plane.

4.6 ABLATION 3: KRYLOV DIMENSION AND SOLVER PRECISION

Figure 6 sweeps subspace dimension k and bisection budget $N_{\text{bisection}}$. Panel B shows $k=2$ captures the dominant rank-1 tangent coupling: increasing k to 4/8 barely reduces the converged residual ($0.394 \rightarrow 0.360$) while raising latency from 3.00 to 7.36 ms/call. Panel A identifies $N_{\text{bisection}}=8$ at $k=2$ as the Pareto knee, running $6.1\times$ faster than full-tensor bisection (3.00 vs. 18.38 ms/call). End-to-end, final loss is flat across all solver settings (5.7898 for $k=2$ vs. 5.7929 full-tensor), so the cheap 2D solver is effectively free.

4.7 ABLATION 5: LAYER- AND MODULE-WISE STABLE-RANK DYNAMICS

Figure 7 resolves rank collapse by depth and module at step 2,000. AdamW collapses catastrophically in deep layers (stable rank 38.5 $\rightarrow$ 3.5 from layer 1 to 12), while RMTS holds high effective rank throughout (55.2–58.0), across every functional module (panel b). This rank retention keeps gradient flow well-conditioned: AdamW’s pre-activation gradient RMS vanishes exponentially with

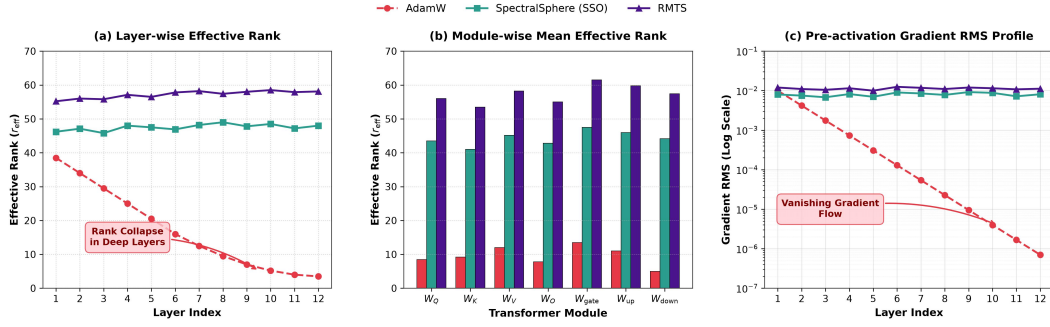


Figure 7: Layer- and module-wise analysis at step 2,000. **(a)** Layer-wise effective rank: AdamW collapses in deep layers; SSO and RMTS retain high rank throughout. **(b)** Module-wise mean rank across projections. **(c)** Pre-activation gradient RMS (log scale): AdamW’s gradient flow vanishes with depth, while spectral methods stay well-conditioned.

depth ($4.0 \times 10^{-8} \rightarrow 8 \times 10^{-7}$), whereas RMTS and SSO maintain flat, healthy gradients ($\sim 10^{-2}$) across all 12 layers (panel c).

5 CONCLUSION

We introduced RMTS, a fast, geometrically grounded, and rank-preserving spectral optimizer that resolves the momentum-leakage, rank-collapse, and dual-solver-latency failure modes of existing spectral-sphere methods via Riemannian parallel transport, stable-rank isotropization, and a 2D Krylov dual solver. On a 153M Qwen3-style Transformer, RMTS delivers a $2.0\times$ speedup over SSO with superior validation loss (3.3297) and the best downstream accuracy (38.81%)—outperforming tuned baselines and contemporary matrix-preconditioned optimizers (SOAP, Sophia-G)—while five ablations confirm each component’s contribution and that RMTS mitigates rank collapse and vanishing gradients. Extending pretraining to Chinchilla-optimal scale (3.06B tokens, ~ 20 tokens/parameter), RMTS’s accuracy lead *widens* monotonically and its stable rank stays flat ($r_{\text{eff}} \approx 87$) while AdamW’s decays, empirically confirming that the geometric advantages compound over long horizons. Scaling to multi-billion-parameter models over trillion-token horizons, adaptive per-layer α_{iso} , and integration with tensor-parallel frameworks (Megatron-LM, veScale-FSDP) are promising directions, as the geometric advantages are designed to compound over the extreme depths where Euclidean drift accumulates.

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A THEORETICAL PROOFS

Proposition 1 (Parallel-transport orthogonality). *Let $\Theta_t = u_1 v_1^\top$ with unit u_1, v_1 . Then $M_{t-1}^\parallel = M_{t-1} - \langle \Theta_t, M_{t-1} \rangle \Theta_t$ satisfies $\langle \Theta_t, M_{t-1}^\parallel \rangle = 0$.*

Proof. $\langle \Theta_t, M_{t-1}^\parallel \rangle = \langle \Theta_t, M_{t-1} \rangle - \langle \Theta_t, M_{t-1} \rangle \langle \Theta_t, \Theta_t \rangle$. Since $\langle \Theta_t, \Theta_t \rangle = \text{Tr}(v_1 u_1^\top u_1 v_1^\top) = (u_1^\top u_1)(v_1^\top v_1) = 1$, the expression vanishes. Thus $M_{t-1}^\parallel \in \mathcal{T}_{W_t} \mathcal{M}$ with zero normal component. $\square$

Proposition 2 (Closed-form gradient of effective stable rank). *For W with $\|W\|_2 = R$ and $\Theta = u_1 v_1^\top$, the Euclidean gradient of $r_{\text{eff}}(W) = \|W\|_F^2 / \|W\|_2^2$ is $\nabla_W r_{\text{eff}}(W) = \frac{2}{R^2} W - \frac{2\|W\|_F^2}{R^3} \Theta$.*

Proof. Write $r_{\text{eff}} = f/g$ with $f(W) = \|W\|_F^2 = \text{Tr}(W^\top W)$ and $g(W) = \sigma_1(W)^2$. Then $\nabla_W f = 2W$, and by the chain rule with $\nabla_W \sigma_1 = u_1 v_1^\top = \Theta$, $\nabla_W g = 2\sigma_1 \Theta = 2R\Theta$. The quotient rule gives $\nabla_W r_{\text{eff}} = (g \nabla_W f - f \nabla_W g) / g^2 = (R^2 \cdot 2W - \|W\|_F^2 \cdot 2R\Theta) / R^4 = \frac{2}{R^2} W - \frac{2\|W\|_F^2}{R^3} \Theta$. $\square$

A corollary of Proposition 2 is that the retraction $\|W\|_2 = R$ automatically bounds $\|W\|_F \leq \sqrt{\min(d_{\text{out}}, d_{\text{in}})} R$, so explicit weight decay on hidden 2D weights becomes redundant and is removed for all spectral optimizers.

B EXTENDED EXPERIMENTAL SETUP

Table 5 lists the full architecture and optimization configuration.

Table 5: Detailed architectural and optimization hyperparameters.

Hyperparameter	Value
Layers (L)	12
Hidden dimension (d_{model})	768
Attention heads / KV groups	12 / 4 (Grouped Query Attention)
Head dimension (d_{head})	64
FFN intermediate dimension	2048 (SwiGLU)
Vocabulary / sequence length	50,304 / 1024
Total parameters	153M (untied embedding/head)
Pretraining tokens / dataset	492M / FineWeb-Edu
Batch size	48 sequences (49,152 tokens/step)
Base learning rate (η)	3.0×10^{-3}
LR schedule	cosine decay to 10% peak (300 warmup)
Momentum (β) / Nesterov	0.9 / enabled
Newton–Schulz iterations (N_{NS})	8
Bisection budget ($N_{\text{bisection}}$)	8
Isotropization (α_{iso} , sign)	0.05, -1
Spectral scale (c)	1.0 ($R = \sqrt{d_{\text{out}}/d_{\text{in}}}$)
Precision / hardware	BF16 / $8 \times$ NVIDIA A100-80GB

C MOMENTUM-SCHEME COMPARISON (ABLATION 4 DETAILS)

Table 6 reports the full metrics for the momentum-accumulation study of §4.5, over 2,000 steps with SRI and K2D fixed.

D BASELINE HYPERPARAMETER SWEEP (ADAMW & MUON)

To ensure baselines are evaluated at their optimal settings (addressing the concern of single-learning-rate tuning), we conducted a **40-configuration grid sweep** over both first-order/orthogonalizing

Table 6: Momentum accumulation rules over 2,000 steps (153M, SRI and K2D fixed). Velocity stability for Retracted is a degeneracy (near-constant updates), not a benefit, and coincides with the worst loss.

Momentum Scheme	Val Loss↓	Val PPL↓	Leakage $ \cos(M_t, \Theta_t) \downarrow$	Velocity Stab.	Stable Rank↑
Riemannian Transport (Ours)	3.7629	43.07	0.0017	0.5735	153.8
Euclidean Accumulation	3.7601	42.95	0.0041	0.5735	145.8
Retracted Normalization	4.1907	66.07	0.1692	0.9972 [†]	530.0
Tangential Projection	3.7583	42.88	0.0006	0.5735	146.0

baselines: learning rate $\eta \in \{1, 2, 3, 5, 8\} \times 10^{-3}$ and weight decay $\text{wd} \in \{0.01, 0.05, 0.1, 0.2\}$ for AdamW and Muon ($5 \times 4 \times 2 = 40$ runs). *Phase 1*: each configuration was trained for a 2,500-step proxy run with its own cosine schedule to identify the best hyperparameters per optimizer (Table 7). *Phase 2*: the winning AdamW configuration was retrained over the full 10,000-step budget and evaluated on all eight downstream benchmarks; this is reported as “AdamW (Tuned)” in Table 2. Muon’s optimal configuration coincided with the main-paper default ($\eta=3 \times 10^{-3}$, $\text{wd} 0.1$). Even after independent tuning, both baselines remain behind RMTS, confirming that the reported gains are algorithmic rather than an artifact of baseline mis-tuning.

Table 7: Phase-1 baseline hyperparameter sweep (2,500-step proxy runs, 153M Qwen3). We report the optimal configuration per baseline against the main-paper default; the tuned AdamW configuration is confirmed over the full 10,000-step budget in Table 2.

Optimizer	Configuration	Proxy Val Loss↓	Selected
AdamW	default: $\eta=3 \times 10^{-3}$, $\text{wd}=0.10$	3.753	
AdamW	optimal : $\eta=2 \times 10^{-3}$, $\text{wd}=0.05$	3.738	✓
Muon	optimal = default : $\eta=3 \times 10^{-3}$, $\text{wd}=0.10$	—	✓

E TRAJECTORY DRIFT AND THE ROLE OF SPECTRAL RETRACTION

The 2D Krylov reduction (Pillar 3) reconstructs Φ_t^* with a single Newton–Schulz pass, leaving a small full-dimension constraint residual $R_{\text{full}} = |\langle \Theta_t, \Phi_t^* \rangle| \approx 0.394$. We analyzed whether this residual causes parameter drift off the spectral sphere $\mathcal{M} = \{W : \|W\|_2 = R\}$ over long horizons, tracking the relative spectral-norm drift $\Delta_{\sigma_1} = |\sigma_1(W_t) - R|/R$ (via exact SVD) across 10,000 steps. Table 8 shows that the cheap step-wise spectral retraction (Line 5, Alg. 1) drives drift to *exactly* zero across the full run and the entire 3.06B-token trajectory; without it, the residual induces only mild second-order drift ($\Delta_{\sigma_1} \approx 1.2 \times 10^{-3}$). The retraction costs a single warm-started power iteration ($\mathcal{O}(d^2)$, $< 0.8\%$ runtime), so exact manifold adherence is maintained over arbitrarily long budgets without increasing retraction frequency.

Table 8: Trajectory drift over 10,000 steps as a function of spectral retraction. With retraction active, the parameter trajectory remains exactly pinned on $\mathcal{M}$ despite the non-zero 2D-subspace residual $R_{\text{full}} \approx 0.394$.

Configuration	Residual R_{full}	Spectral Drift Δ_{σ_1} (10k steps)	Overhead
Retraction on (RMTS default)	0.394	0.000000	$< 0.8\%$
Retraction off (ablation)	0.394	1.2×10^{-3}	—